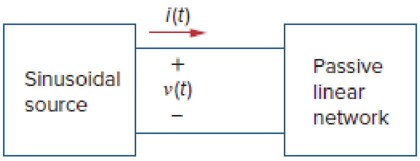
Chapter 11, Problem 1PP

Problem

Calculate the instantaneous power and average power absorbed by the passive linear network of Fig. 11.1 if

$$v(t) = 330\cos(10t + 20^\circ)\text{ V} \qquad \text{and} \qquad i(t) = 33\sin(10t + 60^\circ)\text{ A}$$

Figure 11.1: Sinusoidal source and passive linear circuit.



Step-by-step solution

Step 1 of 3

Refer to Figure 11.1 in the textbook for linear passive network.

Consider the following data:

The sinusoidal expression for voltage is, $v(t) = 330\cos(10t + 20^\circ)\text{ V}$

The sinusoidal expression for current is, $i(t) = 33\sin(10t + 60^\circ)\text{ A}$

Write the current expression in terms of standard cosine function.

$$\begin{aligned} i(t) &= 33\sin(10t + 60^\circ)\text{ A} \\ &= 33\cos(10t + 60 - 90^\circ)\text{ A} \\ &= 33\cos(10t - 30^\circ)\text{ A} \end{aligned}$$

Step 2 of 3

Calculate the instantaneous power p absorbed by the passive linear network.

$$\begin{aligned} p &= vi \\ &= \left[330\cos(10t + 20^\circ)\right]\left[33\sin(10t + 60^\circ)\right] \\ &= (330)(33)\cos(10t + 20^\circ)\sin(10t + 60^\circ) \\ &= 10890\cos(10t + 20^\circ)\sin(10t + 60^\circ) \end{aligned}$$

Apply the following trigonometry identity to the equation.

$$\cos A \sin B = \frac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$$

The following is the instantaneous power p:

$$\begin{aligned} p &= \frac{10890}{2}\left[\sin(10t + 20^\circ + 10t + 60^\circ) - \sin(10t + 20^\circ - 10t - 60^\circ)\right] \\ &= 5445\left[\sin(20t + 80^\circ) - \sin(-40^\circ)\right] \\ &= 5445\left[\sin(20t + 80^\circ) + 0.643\right] \\ &= 3.5 + 5.445\sin(20t + 80^\circ)\text{ kW} \end{aligned}$$

$$\begin{aligned} &= 3.5 + 5.445\cos(20t + 80^\circ - 90^\circ)\text{ kW} \\ &= 3.5 + 5.445\cos(20t - 10^\circ)\text{ kW} \end{aligned}$$

Therefore, the value instantaneous power p absorbed by the passive linear network is $\boxed{3.5 + 5.445\cos(20t - 10^\circ)\text{ kW}}$.

Write the expression for voltage.

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ V}$$
$$v(t) = 330 \cos(10t + 20°) \text{ V}$$

Write the expression for current.

$$i(t) = I_m \cos(\omega t + \theta_i) \text{ A}$$
$$i(t) = 33 \cos(10t - 30°) \text{ A}$$

Calculate the average power P absorbed by the passive linear network.

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \\ &= \frac{1}{2}(330)(33)\cos(20° - (-30°)) \\ &= \frac{1}{2}(330)(33)\cos(50°) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2}(330)(33)(0.6427) \\ &= 3499.98 \text{ W} \\ &\simeq 3.5 \text{ kW} \end{aligned}$$

Therefore, the value of average power P absorbed by the passive linear network is $\boxed{3.5 \text{ kW}}$.